

SuDocu+: Refining Summarization Intent Through Natural Language

Aryan Deshpande*, Vasileios Vittis *, Peter Haas *, Azza Abouzied*, Alexandra Meliou *

* University of Massachusetts, Amherst
* New York University, Abu Dhabi

Background

How do I **summarize text** according to my **own preferences**?



Give me the **interesting bits** of the latest presidential address.



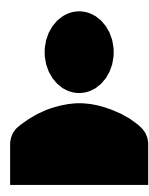
Show me the **exciting aspects** from this travel guide.



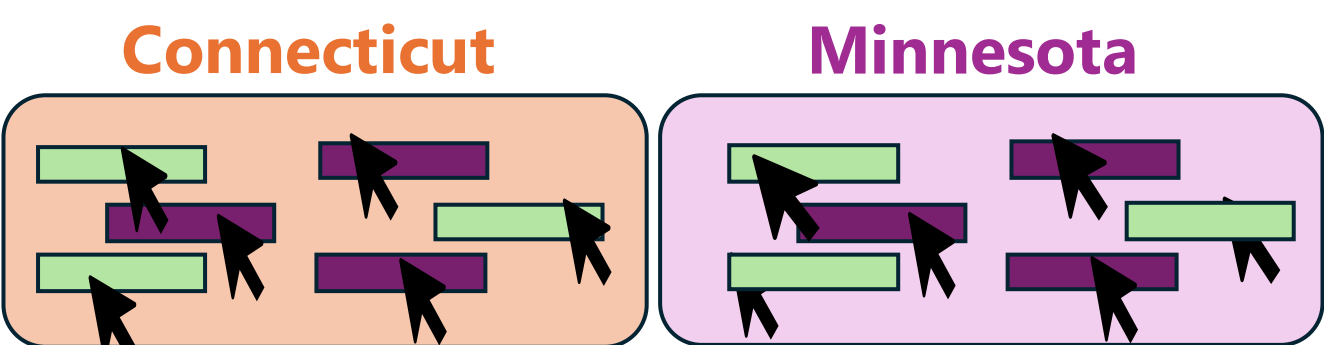
Summary Refinement Flow

SuDocu (original):

Make the summary more **politics**-focused, not just **climate**.



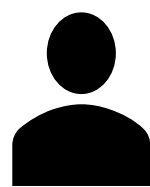
Requires user to create brand-new example summaries from scratch on every refinement.



Manual clicks get very cumbersome over time for user!

SuDocu+ (proposed):

Make the summary more **politics**-focused, not just **climate**.



User types in exactly what they want in natural language just once; system updates **package query** based on request.

SUM(T_{climate}) BETWEEN 3.4 AND 5.1
1.5 AND 2.2

SUM(T_{politics}) BETWEEN 0.4 AND 0.6
1.7 AND 2.9

Way less hassle for user!

Motivation

Problem: Example-based summarization systems provide optimization-based control, but do not support quick refinement that users get from conversational AI at scale.

What if I'm not satisfied with the initial summary?

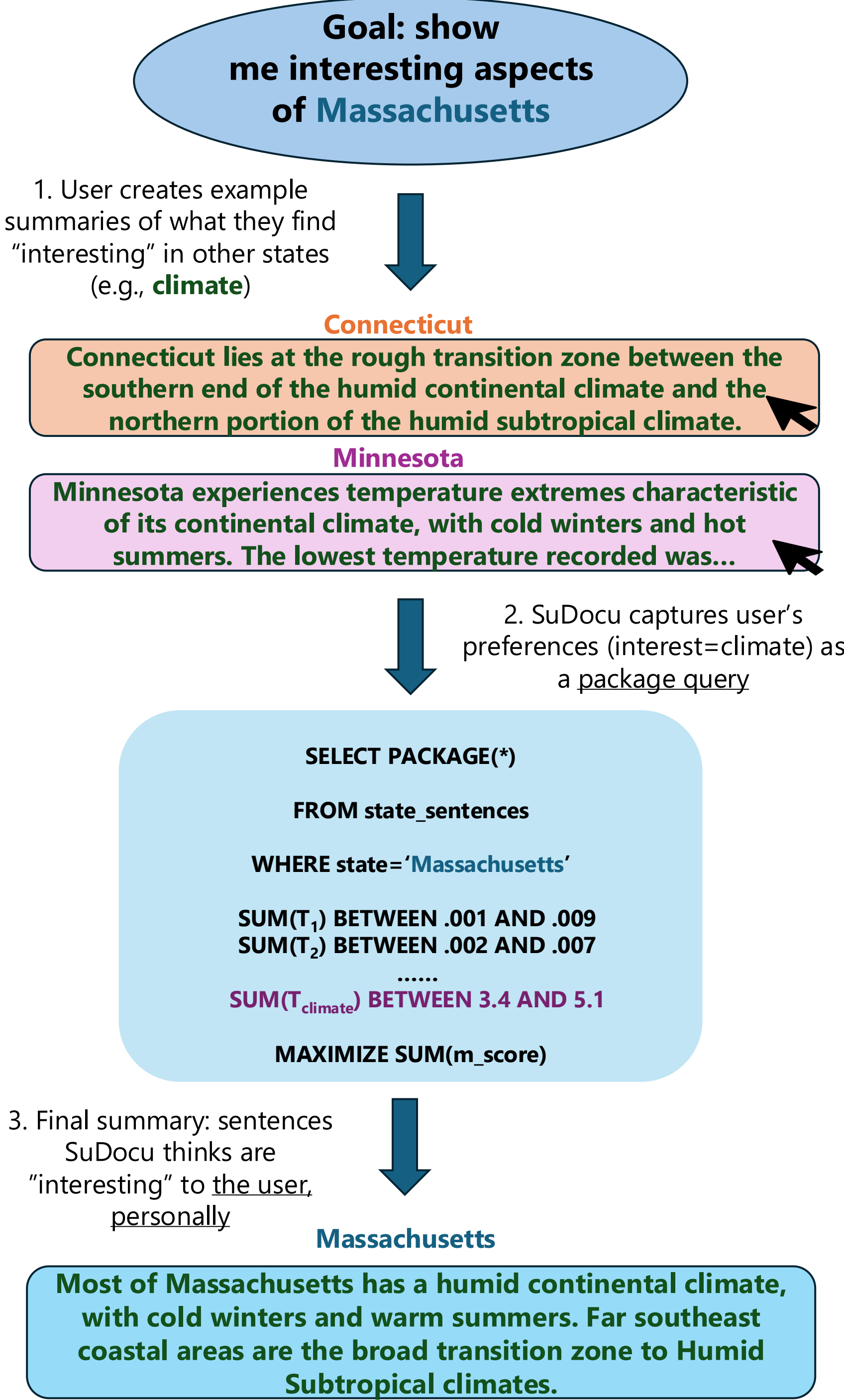
Will I have to make brand-new example summaries?

Let's extend SuDocu to support accurate summary refinement through natural conversation.

Today's users already engage with conversational AI at a massive scale. According to Business Insider, ChatGPT alone had over **700 million users** as of July 2025 [1]. Research also shows that **iteratively refining** AI responses often outperforms one-shot prompting [2]. One notable flaw of AI systems today is their tendency to **hallucinate** [3]; such considerations at a massive scale reinforce the need for systems like SuDocu to meet the demands of the modern user.

SuDocu (Summarizing Documents by Example):

An existing subjective summarization system that works in the following way:



Evaluation of Refinement for Balanced Topic Coverage

	Original	Hybrid 50-50	ChatGPT
Topic 1 (politics)	8.7	2.3	42.2
Topic 2 (education)	8.6	8.3	25.0
Topic 3 (nature)	9.4	2.0	63.5
Topic 4 (urbanization)	18.2	20.0	24.3

Table 1: Average Ranks of Sentences in Desired Summary Topics

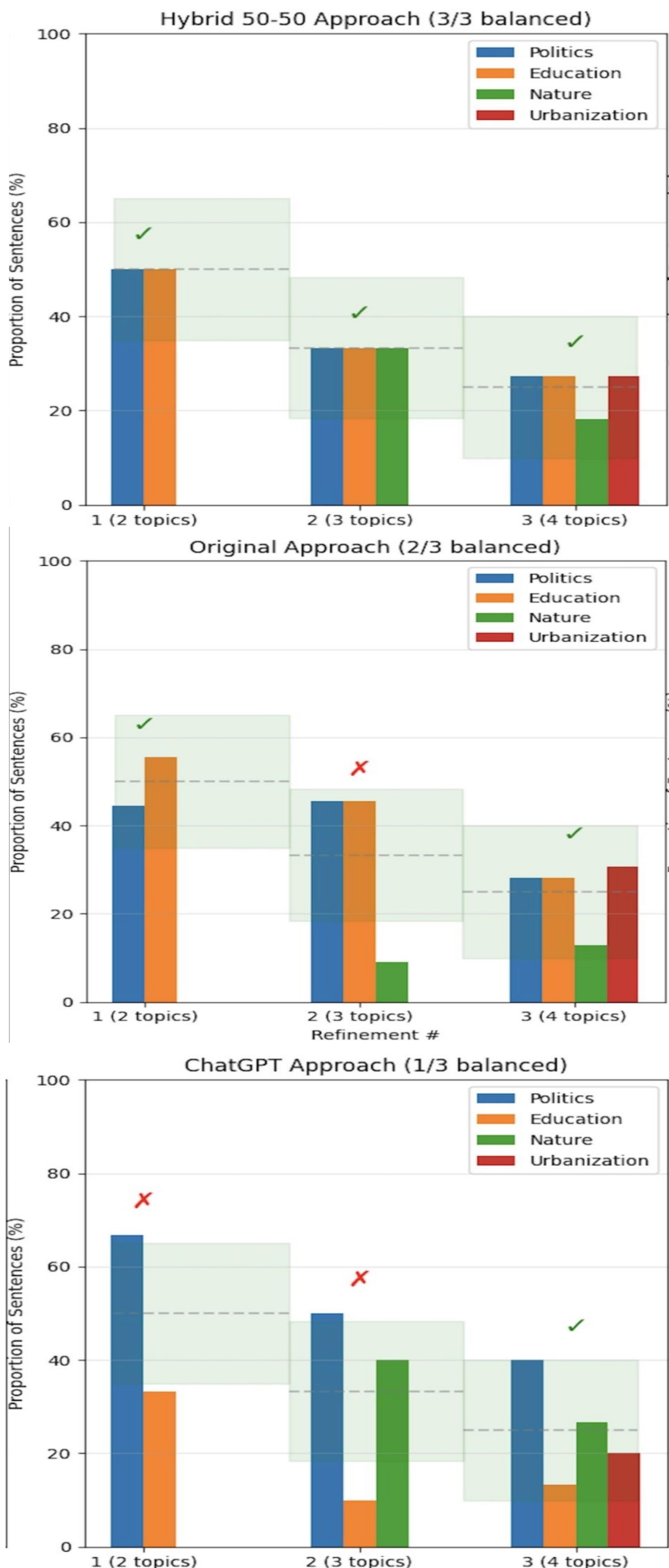


Figure 1: Topic Distributions of Original, Hybrid 50-50, and ChatGPT approaches

Future Work

How do we effectively process user input?
➔Light LLM or classifier model

How can we account for a wide range of refinements?
➔Improved latent topic distribution

References

- [1] Sor, Jennifer. "ChatGPT is now being used by 10% of the world's adult population". *Business Insider*, 9 October 2025.
- [2] Qingquan, T., Feng, R., Bin, Z. et al. Iteratively refined ChatGPT outperforms clinical mentors in generating high-quality interprofessional education clinical scenarios: a comparative study. *BMC Med Educ* 25, 845 (2025). <https://doi.org/10.1186/s12908-025-07414-1>
- [3] Lei Huang, Weijiang Yu, Weitao Mo, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihus Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2025. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Trans. Inf. Syst.* 43, 2, Article 42 (March 2025), 55 pages. <https://doi.org/10.1145/3703155>
- [4] Anna Fariha, Matteo Brucato, Peter J. Haas, and Alexandra Meliou. Sudocu: Summarizing documents by example. *Proc. VLDB Endow.*, 13(12):2861–2864, 2020. doi: 10.14778/3415478.3415494. URL <http://www.vldb.org/pvldb/vol13/p2861-fariha.pdf>.